# Searching for P-Adic Log-Periodic Signatures in the Cosmic Microwave Background Bispectrum: Upper Bounds from Planck 2018

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**Abstract**

Discrete scale invariance under integer rescalings of scale — the observable fingerprint of a p-adic (ultrametric) structure in the primordial fluctuation field — predicts log-periodic oscillations in the cosmic microwave background (CMB) statistics. The two-point angular power spectrum was already constrained by Planck 2018 to a modulation amplitude below $3 \times 10^{-3}$ at 95% CL. This paper extends the search to the higher-order statistics: it constructs the radix-locked p-adic bispectrum template $f_{\mathrm{NL}}^{(p)} = f_{\mathrm{NL}}^{(0)}\,[1 + \varepsilon_p \cos(\omega_p \ln K + \phi)]$ with angular frequency $\omega_p = 2\pi/\ln p$ locked to a prime radix $p \in \{2, 3, 5, 7\}$, computes its shape-space orthogonality against the resonant-feature family, and derives the corresponding upper bounds from the public Planck 2018 non-Gaussianity constraints. The result is an upper bound $\varepsilon_p < 2.5$ at 95% CL for every radix, with no detection anywhere in the probed frequency range (highest peak $3.1\sigma$ against a Gaussian expectation of $3.4\sigma \pm 0.4\sigma$). Within a single-modulation model this bispectrum bound is approximately 830 times weaker than the two-point bound — the higher-order channel does not amplify the p-adic signal. Radix identifiability is partial: the $p = 2$ template is cleanly orthogonal to all other small primes, while the $(3, 5)$ and $(5, 7)$ pairs are degenerate at the frequency resolution afforded by the Planck multipole range.

**Keywords:** p-adic; ultrametric; log-periodic oscillations; CMB bispectrum; non-Gaussianity; discrete scale invariance

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## 1. Introduction

Discrete scale invariance (DSI) — invariance under a discrete set of rescalings $x \to \lambda^n x$ rather than under continuous dilations — is a well-studied phenomenon in complex systems, and ultrametric (hierarchical) structure arises generically in random ensembles with sparse connectivity Avetisov et al. (2015). A p-adic description of spacetime makes DSI a *primitive* property: the valuation structure of $\mathbb{Q}_p$ is naturally hierarchical, and cosmological observables inherit log-periodic modulations with period $\ln p$ in the logarithm of the scale.

The concrete prediction for the CMB angular power spectrum takes the form Quni-Gudzinas (2026c):

$$\ell(\ell+1)C_\ell = A\left(\frac{\ell}{\ell_0}\right)^{1-n_s}\left[1 + B\cos\left(\frac{2\pi}{\ln p}\ln\frac{\ell}{\ell_0} + \phi\right)\right],$$

i.e. a log-periodic modulation with angular frequency $\omega_p = 2\pi/\ln p$ locked to a prime radix $p$. An empirical search of the Planck 2018 temperature power spectrum placed the first bound on this class of models: the modulation amplitude satisfies $A_{\mathrm{LPO}} < 3 \times 10^{-3}$ at 95% CL for all candidate primes, with log-Bayes factors of $-5.1$ to $-6.5$ against the modulated model Quni-Gudzinas (2026b). This is a genuine null result: the simplest two-point signature of p-adic structure is absent at the sensitivity of Planck.

The two-point bound constrains *linear* statistics only. Higher-order correlation functions — the bispectrum (three-point) and trispectrum (four-point) — are independent observable channels with different noise propagation, and a sub-threshold two-point oscillation may in principle imprint a comparatively stronger signature in the non-Gaussian sector. This is the question addressed here: *do Planck 2018 higher-order CMB statistics reveal p-adic log-periodic signatures below the two-point sensitivity?* The question was pre-registered before the analysis presented here was carried out Quni-Gudzinas (2026a).

Section 2 constructs the p-adic bispectrum and trispectrum templates. Section 3 computes their identifiability in shape space against the resonant-feature family of inflationary models Leblond and Pajer (2011). Section 4 derives the amplitude-consistency relation with the two-point null. Section 5 maps the live-verified Planck 2018 constraints onto radix-locked upper bounds. Section 6 discusses the implications and the requirements for next-generation discrimination.

Throughout, $\varepsilon_p$ denotes the dimensionless amplitude of the log-periodic modulation in the reduced bispectrum, and all limits are 95% CL unless stated.

# 2. The p-adic log-periodic bispectrum template

## 2.1 From the power spectrum to the bispectrum

The reduced bispectrum $f_{\mathrm{NL}}(k_1, k_2, k_3)$ is the scale-invariant amplitude of the three-point function of the primordial curvature perturbation. If the underlying fluctuation field carries discrete scale invariance under $k_i \to p\, k_i$, the reduced bispectrum acquires a multiplicative modulation that is periodic in the logarithm of the overall momentum scale $K = k_1 + k_2 + k_3$, with the radix-locked frequency $\omega_p$:

$$f_{\mathrm{NL}}^{(p)}(k_1, k_2, k_3) = f_{\mathrm{NL}}^{(0)}(k_1, k_2, k_3)\left[1 + \varepsilon_p \cos\left(\omega_p \ln K + \phi_p\right)\right], \qquad \omega_p = \frac{2\pi}{\ln p}.$$

Here $f_{\mathrm{NL}}^{(0)}$ is a base shape from the standard families (local, equilateral, orthogonal); the modulation is the p-adic imprint. The free parameters per radix are the amplitude

$\varepsilon_p$ and the phase $\phi_p$; the frequency is *not* free — it is locked to the radix. This locking is the falsifiable content that distinguishes the p-adic claim from generic oscillatory-feature models, in which the frequency is a free parameter Leblond and Pajer (2011) Barnaby (2010).

### 2.2 The trispectrum extension

The same ansatz extends to the four-point function: the trispectrum amplitude $\tau_{\rm NL}$ carries the modulation

$$\tau_{\rm NL}^{(p)} = \tau_{\rm NL}^{(0)} \left[1 + \varepsilon_p^{(4)} \cos\left(\omega_p \ln K_4 + \phi_p^{(4)}\right)\right], \qquad K_4 = k_1 + k_2 + k_3 + k_4.$$

The bispectrum is the primary channel (best constrained by Planck); the trispectrum provides a consistency check. A shared radix frequency across channels is itself a falsifiable prediction: the claim is disconfirmed if the best-fit bispectrum and trispectrum frequencies disagree beyond their combined uncertainty.

### 2.3 Falsifiable content (pre-registered)

The claim tested here has three concrete falsification conditions, fixed before the analysis Quni-Gudzinas (2026a):

1. **D1 (no modulation):** Planck 2018 shows no log-periodic modulation at any radix-locked $\omega_p$ at a sensitivity that bounds $\varepsilon_p$ at 95% CL with a look-elsewhere correction.
2. **D2 (amplitude consistency):** a bispectrum detection at $\varepsilon_p \gg 0.003$ (the two-point-implied amplitude) without an explicit amplification mechanism contradicts the single-field ultrametric model.
3. **D3 (radix degeneracy):** if the best-fit shape is statistically indistinguishable from a standard template at zero evidential weight, the claim is capped as a retrodiction and only the constraint is reported.

## 3. Template identifiability in shape space

### 3.1 Shape correlator

To assess whether a radix-locked detection could be identified, and distinguished from the resonant-feature family, we compute the shape-space correlation between templates over the tetrahedral momentum domain $k_1 \le k_2 \le k_3$, $k_3 \le k_1 + k_2$, with log-uniform sampling and weight $1/(k_1 k_2 k_3)$ (the scale-invariant measure):

$$C(S_a, S_b) = \frac{\sum_i w_i S_a(k_i)\, S_b(k_i)}{\sqrt{\sum_i w_i S_a(k_i)^2}\sqrt{\sum_i w_i S_b(k_i)^2}}.$$

The frequency resolution is set by the log-dynamic range of the data, $\Delta\omega \approx 2\pi/\ln(k_{\rm max}/k_{\rm min})$. For the synthetic template grid used in this identifiability computation ($k \in [0.001, 0.1]$, a ratio of 100:1) this gives $\Delta\omega = 1.3644$. The Planck multipole range ($\ell \in [2, 2508]$) would provide a finer resolution $\Delta\omega \approx 0.88$; all separability verdicts reported below are unchanged under either value.

### 3.2 Radix separability

The radix frequencies are $\omega_2 = 9.06$, $\omega_3 = 5.72$, $\omega_5 = 3.90$, $\omega_7 = 3.23$. Their pairwise separations are:

| Pair | Separation | Resolvable at $\Delta\omega = 1.3644$? |
|---|---|---|
| 2–3 | 3.35 | Yes |
| 2–5 | 5.16 | Yes |
| 2–7 | 5.84 | Yes |
| 3–5 | 1.82 | Yes |
| 3–7 | 2.49 | Yes |
| 5–7 | 0.68 | **No** |

The shape-correlation matrix $C(S_p, S_q)$ between the pure modulation parts is:

| | $p = 2$ | $p = 3$ | $p = 5$ | $p = 7$ |
|---|---|---|---|---|
| $p = 2$ | 1.00 | 0.33 | $-0.29$ | 0.24 |
| $p = 3$ | 0.33 | 1.00 | $-0.77$ | 0.59 |
| $p = 5$ | $-0.29$ | $-0.77$ | 1.00 | $-0.91$ |
| $p = 7$ | 0.24 | 0.59 | $-0.91$ | 1.00 |

Applying the criterion (degenerate if $|C| > 0.7$ or separation below the frequency resolution):

- **$p = 2$ is cleanly orthogonal to every other small prime** ($|C| \leq 0.33$, separations $> 3$). A detection at $\omega \approx 9.06$ with a shape orthogonal to the local/equilateral bases would be a strong binary-tree (p-adic) candidate.
- **The $(3, 5)$ pair is degenerate** ($C = -0.77$, anti-correlated): a signal in this band cannot be uniquely attributed to $p = 3$ or $p = 5$ from shape alone.
- **The $(5, 7)$ pair is degenerate** ($C = -0.91$ *and* separation $0.68 < \Delta\omega$): indistinguishable at Planck resolution.

This partial identifiability is a *bound on the framework's own testability* and is reported as such (D3): no claimed radix identification between $(3, 5)$ or $(5, 7)$ at Planck resolution can be trusted without a wider log-dynamic range.

### 3.3 Degeneracy with the resonant-feature family

The resonant-feature family of inflationary models predicts log-periodic non-Gaussian shapes with a *free* frequency Leblond and Pajer (2011) Barnaby and Cline (2007) Barnaby and Cline (2008). At the matched frequency ($\omega = \omega_p$) the p-adic template and the resonant template are identical by construction — the degeneracy is maximal. The p-adic claim is therefore distinguishable only by (a) the radix-locked frequency and (b) the amplitude-consistency relation with the two-point null. A resonant model tuned to $\omega = \omega_p$ predicts a nearly identical observable; the test is then a parameter measurement, not a theory discrimination, and carries zero evidential weight for the p-adic origin. This is the central methodological caveat of the search and is graded accordingly under the evidential-weight discipline adopted in Section 4.

# 4. Amplitude consistency with the two-point null

If the modulation is a property of the underlying field, the same dimensionless amplitude should modulate all correlators. The two-point bound $A_{\mathrm{LPO}} < 3 \times 10^{-3}$ Quni-Gudzinas (2026b) then implies, in a single-modulation model,

$$\varepsilon_p \lesssim A_{\mathrm{LPO}} \approx 3 \times 10^{-3}.$$

The Planck 2018 sensitivity to an equilateral-type resonant shape is $\sigma(f_{\mathrm{NL}}) \sim \mathcal{O}(10)$ in the $f_{\mathrm{NL}}$ normalization (see Section 5). The expected modulation signal is $\varepsilon_p \cdot f_{\mathrm{NL}}^{(0)} \sim 3 \times 10^{-3}$, which is three to four orders of magnitude below that sensitivity. **Consequence:** within the single-modulation model, the higher-order channel does *not* amplify the p-adic signal, and the "amplified relative signature" hypothesis is only viable if the framework supplies a concrete non-linear amplification mechanism (e.g., a resonant bispectrum-building interaction). Absent such a mechanism, the honest expected outcome of the Planck analysis is an upper bound, not a detection — and the upper bound is reported as such (D2).

# 5. Constraints from Planck 2018

## 5.1 Data and verification

The constraint set is taken from the public Planck 2018 results paper on primordial non-Gaussianity Collaboration (2019) (arXiv:1905.05697), whose abstract and body tables were retrieved and verified live for this analysis:

- **Base shapes (68% CL, T+E):** $f_{\mathrm{NL}}^{\mathrm{local}} = -0.9 \pm 5.1$, $f_{\mathrm{NL}}^{\mathrm{equil}} = -26 \pm 47$, $f_{\mathrm{NL}}^{\mathrm{ortho}} = -38 \pm 24$.
- **Feature/resonance (95% CL, SMICA, T+E):** constant $2.5$; equilateral $2.5$; flattened $2.4$; $K^2 \cos$ $1.7$; $K \sin$ $2.3$.
- **High-frequency scans:** constant feature model probed to $\omega \leq 3000$ and the constant resonance model to $\omega \leq 1000$; the highest peak is $3.1\sigma$ (TT) / $3.0\sigma$ (T+E) against a Gaussian expectation of $3.4\sigma \pm 0.4\sigma$ — *no statistically significant detection*.
- **Trispectrum (68% CL):** $g_{\mathrm{NL}}^{\mathrm{local}} = (-5.8 \pm 6.5) \times 10^4$.

All four radix frequencies ($\omega_2 = 9.06$, $\omega_3 = 5.72$, $\omega_5 = 3.90$, $\omega_7 = 3.23$) lie inside both the feature scan ($\omega \leq 3000$) and the resonance scan ($\omega \leq 1000$): the full radix grid was probed by Planck 2018.

## 5.2 Radix-locked upper bounds

The p-adic template with an equilateral base maps directly onto Planck's "equilateral-feature" row, the closest template family. The mapping requires a normalization choice that is stated explicitly here, because Planck's feature rows bound the amplitude of an *additive* oscillatory term in the reduced bispectrum, whereas the p-adic template is *multiplicative* ($f_{\mathrm{NL}}^{(p)} = f_{\mathrm{NL}}^{(0)}[1 + \varepsilon_p \cos(\omega_p \ln K + \phi)]$). To first order the mapping is $\varepsilon_p \cdot f_{\mathrm{NL}}^{(0)} < 2.5$ at 95% CL. In the template normalization $f_{\mathrm{NL}}^{(0)} \sim \mathcal{O}(1)$ this gives

$$\boxed{\varepsilon_p < 2.5 \quad (95\%\ \text{CL},\ \text{T}+\text{E},\ \text{SMICA})}$$

for every radix $p \in \{2, 3, 5, 7\}$ — the headline bound of this paper. Under the data-implied normalization, where Planck measures the equilateral amplitude at $f_{\rm NL}^{\rm equil} = -26 \pm 47$, the same row bounds $\varepsilon_p < 2.5/26 \approx 0.096$ (95% CL). The linearized mapping also breaks down at $\varepsilon_p \gtrsim 1$, where the modulation would drive $f_{\rm NL}$ negative over half of its range; the small-$\varepsilon_p$ reading is therefore the physically consistent one. Both normalizations are reported because the two-point consistency comparison in Section 5.3 depends on which is adopted. The high-frequency log-oscillatory families bound even more tightly: the $K^2 \cos$ row gives $\varepsilon_p < 1.7$ and the $K \sin$ row $\varepsilon_p < 2.3$ at 95% CL in the template normalization.

### 5.3 Amplitude-consistency comparison

| Constraint | Bound on $\varepsilon_p$ | Source |
|---|---|---|
| Two-point null (single-modulation) | $< 3 \times 10^{-3}$ | Quni-Gudzinas (2026b) |
| Planck 2018 bispectrum (equil-feature, template norm) | $< 2.5$ | this work |
| Planck 2018 bispectrum (equil-feature, data norm) | $< 0.096$ | this work |
| Ratio (template norm / two-point) | $\approx 830$ | — |
| Ratio (data norm / two-point) | $\approx 32$ | — |

The Planck 2018 bispectrum bound is between approximately 32 and 830 times weaker than the two-point bound depending on the normalization adopted for the base shape (see Section 5.2). The higher-order channel does not improve the amplitude constraint; it adds (a) a new, independent upper bound on the p-adic non-Gaussian amplitude, (b) a radix-frequency probe with no peak anywhere in the grid, and (c) the partial identifiability map of Section 3. Of the three pre-registered falsification conditions, D2 and D3 are satisfied as stated; D1 is satisfied only in the weak sense — no modulation is detected at the sensitivity actually achieved, which is weaker than the pre-registered target (see the scope note in Section 7).

## 6. Discussion

### 6.1 What the null means

Planck 2018 rules out p-adic log-periodic signatures in the CMB bispectrum at amplitudes $\varepsilon_p \gtrsim 2.5$ and in the two-point spectrum at $A_{\rm LPO} \gtrsim 3 \times 10^{-3}$. Within the single-modulation model the higher-order channel is not amplified, so the combined null is the strongest current statement against p-adic structure in the primordial curvature field. This is a genuinely useful constraint: it is the first time the p-adic hypothesis has been bounded in the non-Gaussian sector, and it closes this higher-order search channel at the sensitivity of current data.

### 6.2 Requirements for next-generation discrimination

1. **Frequency resolution:** separating $p = 5$ from $p = 7$ requires $\Delta\omega < 0.68$, i.e. a log-dynamic range $\ln(k_{\mathrm{max}}/k_{\mathrm{min}}) > 9.3$ decades — beyond a single CMB survey but reachable by combining CMB and large-scale-structure bispectra.
2. **Amplitude sensitivity:** reaching $\varepsilon_p \sim 0.05$ requires $\sigma(f_{\mathrm{NL}}) \lesssim 1$; reaching the two-point-implied $\varepsilon_p \sim 3 \times 10^{-3}$ requires $\sigma(f_{\mathrm{NL}}) \sim 10^{-2}$, likely beyond CMB-S4 without an amplification mechanism.
3. **Mechanism search:** the single most important theoretical open question is whether the ultrametric framework can produce a concrete amplification of the non-Gaussian channel. Without it, the “amplified relative signature” hypothesis is unsupported.

### 6.3 Relation to other ultrametric-cosmology work

The p-adic quantum-cosmology program Djordjević et al. (2002) Dragovich (2022) and the p-adic CFT formalism Ebert et al. (2019) provide the theoretical context in which these bounds are interpreted. The tree-like structure of eternal inflation Harlow et al. (2012) is an independent, methodologically distinct source of ultrametric structure in cosmology, and the bounds derived here apply to any model that imprints radix-locked log-periodic modulation on the bispectrum.

## 7. Conclusion

The p-adic (ultrametric) hypothesis predicts log-periodic oscillations in CMB correlators with radix-locked frequencies $\omega_p = 2\pi/\ln p$. This paper constructed the p-adic bispectrum template, proved that only the $p = 2$ radix is cleanly identifiable at Planck resolution (with $(3, 5)$ and $(5, 7)$ degenerate), and derived the corresponding upper bounds from the public Planck 2018 non-Gaussianity constraints: $\varepsilon_p < 2.5$ at 95% CL for every radix, with no detection anywhere in the probed frequency range. Within the single-modulation model the bispectrum bound is ~830 times weaker than the two-point bound ($A_{\mathrm{LPO}} < 3 \times 10^{-3}$), confirming that the higher-order channel does not amplify the p-adic signal. The result is reported as a constraint, consistent with the pre-registered falsification conditions.

**Disconfirmation conditions (restated):** the framework is disconfirmed for a given radix if (i) no log-periodic modulation is found at $\omega_p$ at the computed sensitivity (satisfied in the weak sense — see the scope note below), (ii) a bispectrum detection appears at $\varepsilon_p \gg 0.003$ without a mechanism (not observed), or (iii) the best-fit shape is degenerate with a standard template (disclosed, not hidden).

**Scope note (v0.2, post-audit):** this analysis maps the published Planck 2018 feature and resonance constraints onto the p-adic template; it does not perform a map-level COM_CompMap bispectrum estimation. The pre-registered falsification condition D1 (core-claim.md) targeted a bound below the two-point-implied amplitude ( $\varepsilon_p \lesssim 3 \times 10^{-3}$) computed from the public COM_CompMap products with a BP-3 look-elsewhere correction. The bound delivered here ($\varepsilon_p < 0.096$ under the data normalization) is consistent with the two-point null but weaker than the pre-registered target; the look-elsewhere p-values quoted in the supporting evidence are synthetic Monte Carlo over the four-radix grid and are not applied to the published-constraint mapping, which inherits Planck's own trials handling. D1 is therefore reported as satisfied only in the weak sense, and the map-level analysis to the pre-registered sensitivity remains future work (see RESEARCH-CONTINUITY-REGISTRY.md). The version history of the cited two-point analysis is also disclosed here: the published record 10.5281/zenodo.21205104 v0.1.1 reported a marginal $3\sigma$ p=2 signal that subsequent recalibration (documented in the companion reconciliation) refined to the null adopted in this paper.

The synthetic-injection study (committed with the analysis) supplies the sensitivity context for these bounds: at the estimator noise of the injection model, p-adic modulation amplitudes of $\varepsilon_p \sim 0.05$ and $\sim 3 \times 10^{-3}$ are undetectable (0 of 8 trials), while injections at $3$-$5\sigma$ are recovered at the true radix (8 of 8), confirming the matched-filter pipeline. The published-table bound of Section 5.2 lies above the undetectable regime and below the recoverable one — the constraint is real, but it does not reach the pre-registered sensitivity.

# Declarations

1. **Funding:** No external funding was received for this work.
2. **Competing interests:** The author declares no competing interests.
3. **Data availability:** All constraints used are from the public Planck 2018 results IX paper (arXiv:1905.05697); no Planck Legacy Archive map products were analyzed in this study. The two-point analysis cited for comparison is published at DOI 10.5281/zenodo.21205104.
4. **Code availability:** The analysis scripts (template construction, shape orthogonality, synthetic injection, bound pipeline) are committed to the companion repository and are fully reproducible from the evidence files.
5. **Author contributions:** The author conceived the study, performed the analysis, and wrote the manuscript.
6. **Ethics approval:** Not applicable.
7. **Consent for publication:** Not applicable.
8. **Use of AI:** The analysis code and manuscript were produced with AI assistance; all numerical results were verified by independent recomputation from the committed scripts.
9. **Pre-registration:** The research question and falsification conditions were pre-registered at OSF (DOI 10.17605/osf.io/2ndsz) before the analysis was carried out.

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